Contents

AIME Number Theory Problem Collection **7**
1983 AIME Problem 6: Powers one away from seven . . . 7
1983 AIME Problem 8: A prime divisor of a central binomial coefficient . . . 7
1983 AIME Problem 12: A diameter with reversed chord digits . . . 7
1984 AIME Problem 2: The smallest zero-eight multiple . . . 7
1984 AIME Problem 14: The last even number without two odd composites . . . 8
1985 AIME Problem 3: An integer complex cube . . . 8
1985 AIME Problem 7: Two perfect powers nineteen apart . . . 8
1985 AIME Problem 13: The largest gcd of adjacent quadratic values . . . 8
1986 AIME Problem 5: A variable divisor and a constant remainder . . . 8
1986 AIME Problem 6: A page counted twice . . . 8
1986 AIME Problem 7: Ranking subsets by ternary value . . . 8
1986 AIME Problem 8: Logarithms of proper divisors . . . 8
1987 AIME Problem 1: Addition without carrying . . . 9
1987 AIME Problem 3: Numbers equal to their proper-divisor product . . . 9
1987 AIME Problem 5: A factored equation in two integer squares . . . 9
1987 AIME Problem 7: LCM triples . . . 9
1987 AIME Problem 8: Exactly one integer in a moving interval . . . 9
1987 AIME Problem 11: The longest consecutive-integer representation . . . 9
1987 AIME Problem 12: An integer just above a cube . . . 9
1988 AIME Problem 2: A digit-sum iteration . . . 9
1988 AIME Problem 5: Divisor probability . . . 9
1988 AIME Problem 8: A functional equation hiding the least common multiple . . . 10
1988 AIME Problem 9: The smallest cube ending in eight-eight-eight . . . 10
1989 AIME Problem 3: A repeated decimal digit forced by divisibility . . . 10
1989 AIME Problem 4: One middle term, two perfect powers . . . 10
1989 AIME Problem 9: Recovering a fifth root with congruences . . . 10
1989 AIME Problem 14: Real integers written in a complex base . . . 10
1990 AIME Problem 1: Ranking nonsquares and noncubes . . . 10
1990 AIME Problem 3: A largest regular-polygon side count . . . 10
1990 AIME Problem 5: The smallest multiple with seventy-five divisors . . . 10
1990 AIME Problem 10: Products of two root-of-unity groups . . . 11
1990 AIME Problem 11: A factorial as a shorter consecutive product . . . 11
1990 AIME Problem 13: Leading nines in powers . . . 11
1991 AIME Problem 1: Symmetric sums with an integer restriction . . . 11
1991 AIME Problem 5: Coprime factor allocations . . . 11
1991 AIME Problem 6: Recovering a number from a floor sum . . . 11
1991 AIME Problem 8: Integer roots via a shifted factorization . . . 11
1991 AIME Problem 13: Sock counts from a probability . . . 11
1991 AIME Problem 15: An attainable vector minimum that is integral . . . 11
1992 AIME Problem 1: Pairing reduced fractions around five . . . 12
1992 AIME Problem 5: Numerators of repeating decimals . . . 12
1992 AIME Problem 6: Consecutive integers without carries . . . 12
1992 AIME Problem 11: The order of two successive reflections . . . 12
1992 AIME Problem 15: Missing factorial trailing-zero counts . . . 12
1993 AIME Problem 4: Equal outer and inner sums . . . 12
1993 AIME Problem 6: One integer as three consecutive sums . . . 12

1993 AIME Problem 9: Colliding labels in a circular walk . . . 12
1994 AIME Problem 1: Indexing squares one above multiples of three . . . 13
1994 AIME Problem 4: A sum of binary logarithm floors . . . 13
1994 AIME Problem 5: A weighted sum over digit strings . . . 13
1994 AIME Problem 7: Lines with only integral circle intersections . . . 13
1994 AIME Problem 10: A prime-cube projection in an integer right triangle . . . 13
1994 AIME Problem 11: Distinct heights of a brick tower . . . 13
1994 AIME Problem 12: The finest square grid within a fence budget . . . 13
1995 AIME Problem 2: A product of roots without solving them . . . 13
1995 AIME Problem 6: Divisors below a square root . . . 14
1995 AIME Problem 8: Two simultaneous integer ratios . . . 14
1995 AIME Problem 10: A composite term in five consecutive residue steps . . . 14
1995 AIME Problem 11: A self-similar cut prism . . . 14
1995 AIME Problem 13: Grouping a sum by rounded fourth roots . . . 14
1996 AIME Problem 2: Even logarithm floors . . . 14
1996 AIME Problem 8: Integer pairs of prescribed harmonic mean . . . 14
1996 AIME Problem 10: An integer angle from a tangent identity . . . 14
1996 AIME Problem 14: Cubes crossed by a space diagonal . . . 14
1997 AIME Problem 1: Differences of squares . . . 15
1997 AIME Problem 3: Concatenation mistaken for multiplication . . . 15
1997 AIME Problem 5: Decimal numbers closest to two sevenths . . . 15
1997 AIME Problem 6: An equilateral attachment and another regular polygon . . . 15
1998 AIME Problem 1: Counting possible LCM partners . . . 15
1998 AIME Problem 5: A four-term cycle of signed triangular numbers . . . 15
1998 AIME Problem 8: The longest subtractive recurrence . . . 15
1998 AIME Problem 14: A sharp bound for a box volume ratio . . . 15
1999 AIME Problem 1: Five primes in arithmetic progression . . . 16
1999 AIME Problem 3: A quadratic square condition with four solutions . . . 16
1999 AIME Problem 5: Possible digit-sum changes . . . 16
1999 AIME Problem 7: Four-position divisor switches . . . 16
1999 AIME Problem 13: Distinct tournament win totals . . . 16
2000 AIME I Problem 1: Zero-free factors of powers of ten . . . 16
2000 AIME I Problem 2: A reflected pentagon . . . 16
2000 AIME I Problem 5: Two urns with a fixed total . . . 16
2000 AIME I Problem 6: A gap between means . . . 17
2000 AIME I Problem 11: Independent prime exponents . . . 17
2000 AIME II Problem 2: Lattice points on a difference of squares . . . 17
2000 AIME II Problem 4: Prescribed odd and even divisor counts . . . 17
2000 AIME II Problem 11: Integral equal legs restrict base directions . . . 17
2000 AIME II Problem 14: Borrowing in factorial base . . . 17
2001 AIME I Problem 1: Two digits dividing their number . . . 17
2001 AIME I Problem 8: Largest number with doubled reinterpreted digits . . . 17
2001 AIME I Problem 11: Two numberings of a rectangular array . . . 18
2001 AIME II Problem 1: Digit adjacency as a directed graph . . . 18
2001 AIME II Problem 5: Triangle-free ordered sets . . . 18
2001 AIME II Problem 10: Counting exponent gaps . . . 18
2001 AIME II Problem 14: Unit roots of a sparse polynomial . . . 18
2002 AIME I Problem 3: Reversed two-digit ages . . . 18
2002 AIME I Problem 4: A telescope followed by a prime-square factorization . . . 18

2002 AIME I Problem 7: Three fractional digits of an enormous power . . . 18
2002 AIME I Problem 8: The first repeated Fibonacci-type endpoint . . . 19
2002 AIME I Problem 9: Three arithmetic progressions partition the integers . . . 19
2002 AIME I Problem 11: Unfolding a reflected ray in a cube . . . 19
2002 AIME I Problem 14: Integer means after deletion . . . 19
2002 AIME II Problem 1: Differences of reversed three-digit integers . . . 19
2002 AIME II Problem 3: An integral geometric progression with a square gap . . . 19
2002 AIME II Problem 5: When a power fails to divide another . . . 19
2002 AIME II Problem 7: A square sum divisible by 200 . . . 19
2002 AIME II Problem 8: The first missing integer quotient . . . 20
2003 AIME I Problem 1: The largest factorial factor . . . 20
2003 AIME I Problem 7: Integral distances in an isosceles triangle . . . 20
2003 AIME I Problem 8: An arithmetic triple followed by a geometric triple . . . 20
2003 AIME I Problem 14: The smallest denominator containing a decimal block . . . 20
2003 AIME II Problem 1: A product equal to six times a sum . . . 20
2003 AIME II Problem 2: Largest distinct-digit multiple of eight . . . 20
2003 AIME II Problem 10: A nonsquare radical sum with fixed difference . . . 20
2003 AIME II Problem 12: Minimum occupancy with percentage constraints . . . 21
2004 AIME I Problem 1: Four descending consecutive digits . . . 21
2004 AIME I Problem 5: Strict daily rates with a larger aggregate score . . . 21
2004 AIME I Problem 8: Regular star polygons . . . 21
2004 AIME I Problem 15: Reverse dynamics and forbidden operation runs . . . 21
2004 AIME II Problem 3: Hidden cubes . . . 21
2004 AIME II Problem 6: Integer shares in a three-stage distribution . . . 21
2004 AIME II Problem 8: Divisor counts as exponent allocations . . . 21
2004 AIME II Problem 10: Two binary ones modulo nine . . . 22
2004 AIME II Problem 14: Distinct digit lengths of a sum . . . 22
2005 AIME I Problem 2: Arithmetic progressions reaching 2005 . . . 22
2005 AIME I Problem 3: Exactly three small proper divisors . . . 22
2005 AIME I Problem 4: A rectangular formation five above a square . . . 22
2005 AIME I Problem 12: Parity changes only at squares . . . 22
2005 AIME II Problem 4: Union of divisor sets . . . 22
2005 AIME II Problem 5: Reciprocal logarithms force powers . . . 22
2005 AIME II Problem 6: Fixed points of a shuffle . . . 23
2005 AIME II Problem 9: A swapped sine-cosine power identity . . . 23
2005 AIME II Problem 13: Integer polynomial values constrain integer roots . . . 23
2006 AIME I Problem 3: Deleting the leading digit . . . 23
2006 AIME I Problem 4: Valuation of a product of factorials . . . 23
2006 AIME I Problem 9: Integral geometric progressions and logarithms . . . 23
2006 AIME I Problem 13: Valuation layers . . . 23
2006 AIME II Problem 3: Three-adic valuation of an odd factorial . . . 23
2006 AIME II Problem 7: Complementary decimal numbers . . . 24
2006 AIME II Problem 13: Five representations by consecutive odd integers . . . 24
2006 AIME II Problem 14: Reciprocal digit frequency . . . 24
2007 AIME I Problem 1: Square multiples of twenty-four . . . 24
2007 AIME I Problem 4: The next planetary alignment . . . 24
2007 AIME I Problem 5: A periodic rounding map . . . 24
2007 AIME I Problem 7: A floor-ceiling logarithm indicator . . . 24
2007 AIME I Problem 11: Rounding square roots in blocks . . . 24

2007 AIME II Problem 2: Divisibility becomes a composition . . . 25
2007 AIME II Problem 5: Grid cells below a rational line . . . 25
2007 AIME II Problem 7: Multiples inside a cube-root interval . . . 25
2007 AIME II Problem 12: The largest term controls a geometric sum . . . 25
2007 AIME II Problem 13: Pascal propagation modulo three . . . 25
2008 AIME I Problem 3: Integer speeds from two distance totals . . . 25
2008 AIME I Problem 4: A shifted difference of squares . . . 25
2008 AIME I Problem 6: Summing an arithmetic row repeatedly . . . 25
2008 AIME I Problem 7: Occupied hundred-blocks . . . 26
2008 AIME II Problem 4: Balanced ternary support . . . 26
2008 AIME II Problem 8: A trigonometric telescope and divisibility . . . 26
2008 AIME II Problem 15: Consecutive cubes and a second square . . . 26
2009 AIME I Problem 1: Extremal three-digit geometric progressions . . . 26
2009 AIME I Problem 6: Integer values of a stepped power function . . . 26
2009 AIME I Problem 7: An exponential recurrence that telescopes . . . 26
2009 AIME I Problem 11: Integral triangle areas on a line . . . 26
2009 AIME I Problem 13: Infinite integrality forces a two-cycle . . . 27
2009 AIME I Problem 14: Minimizing a second moment with fixed first and fourth moments 27
2009 AIME II Problem 4: The smallest winning score . . . 27
2009 AIME II Problem 7: A sum with strictly increasing dyadic denominators . . . 27
2009 AIME II Problem 9: An injection leaves two boundary slices . . . 27
2009 AIME II Problem 11: Exactly fifty logarithmic neighbors . . . 27
2010 AIME I Problem 1: One square divisor in a pair . . . 27
2010 AIME I Problem 2: A product of repeated nines . . . 27
2010 AIME I Problem 5: Equality in a difference-of-squares bound . . . 28
2010 AIME I Problem 8: The smallest disk enclosing floor-defined squares . . . 28
2010 AIME I Problem 10: Digits allowed to carry . . . 28
2010 AIME I Problem 12: A multiplicative two-color threshold . . . 28
2010 AIME I Problem 14: A monotone sum of decimal digit counts . . . 28
2010 AIME II Problem 1: The largest multiple with distinct even digits . . . 28
2010 AIME II Problem 3: Valuation of all pair differences . . . 28
2010 AIME II Problem 6: The first reducible quartic . . . 28
2010 AIME II Problem 10: Quadratics modulo interchange of roots . . . 29
2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter . . . 29
2011 AIME I Problem 5: Sliding sums force periodic residues . . . 29
2011 AIME I Problem 7: Splitting a power into equal smaller powers . . . 29
2011 AIME I Problem 10: Obtuse triangles in a regular polygon . . . 29
2011 AIME I Problem 11: Summing a power-residue orbit without its order . . . 29
2011 AIME I Problem 15: Bounding integer roots of a depressed cubic . . . 29
2011 AIME II Problem 3: Integer angles in arithmetic progression . . . 29
2011 AIME II Problem 14: Permutations respecting residue partitions . . . 30
2011 AIME II Problem 15: Floor equality on three intervals . . . 30
2012 AIME I Problem 1: Divisibility in both digit directions . . . 30
2012 AIME I Problem 4: A shared horse and the first integer meeting point . . . 30
2012 AIME I Problem 5: Consecutive binary numbers with equal weight . . . 30
2012 AIME I Problem 6: A positive imaginary part selects a root-of-unity denominator . . . 30
2012 AIME I Problem 10: Order the squares with a prescribed three-digit suffix . . . 30
2012 AIME I Problem 11: Reachability in a lattice semigroup . . . 31
2012 AIME I Problem 15: Multiplication changes every circular separation . . . 31

2012 AIME II Problem 1: Count lattice points on a positive Diophantine segment 31
2012 AIME II Problem 12: Independent forbidden residues 31
2013 AIME I Problem 2: Uniform digit sums modulo five 31
2013 AIME I Problem 8: A function domain imposes an integer divisor 31
2013 AIME I Problem 10: Conjugate integer roots and distinct cubics 31
2013 AIME I Problem 11: Three remainder conditions after an lcm reduction 31
2013 AIME I Problem 15: Two arithmetic progressions with reordered residues 32
2013 AIME II Problem 2: Nested logarithms impose a power-of-two divisor 32
2013 AIME II Problem 6: The first complete thousand-block between squares 32
2013 AIME II Problem 7: Staff reductions force a unique feasible integer solution 32
2013 AIME II Problem 14: Maximize a remainder by separating quotient ranges 32
2014 AIME I Problem 3: Reduced fractions with fixed numerator-plus-denominator 32
2014 AIME I Problem 6: Two parabolas with a shared axis 32
2014 AIME I Problem 8: A four-digit idempotent 32
2014 AIME II Problem 4: Compatible repeating blocks 33
2014 AIME II Problem 15: Prime multiplication as binary carrying 33
2015 AIME I Problem 3: A prime concealed in a cube 33
2015 AIME I Problem 8: Digit sums determine the carries 33
2015 AIME I Problem 11: Integral isosceles triangles with a fixed incenter distance 33
2015 AIME II Problem 1: Two percentage relations force a least common multiple 33
2015 AIME II Problem 3: A multiple with a prescribed digit sum 33
2015 AIME II Problem 6: Two possible cubics with positive integer roots 33
2015 AIME II Problem 8: An integer inequality bounds a cubic quotient 34
2015 AIME II Problem 13: Negative partial sums of successive sines 34
2016 AIME I Problem 5: Two arithmetic totals share a day count 34
2016 AIME I Problem 7: Real values of a complex radical expression 34
2016 AIME I Problem 10: Alternating geometric and arithmetic triples 34
2016 AIME I Problem 12: The first value with four prime factors 34
2016 AIME II Problem 5: An infinite chain of right-triangle altitudes 34
2016 AIME II Problem 7: Three nested squares with two integer areas 34
2016 AIME II Problem 8: Three distinct factors of a product 35
2016 AIME II Problem 9: An arithmetic sequence plus a geometric sequence 35
2016 AIME II Problem 11: Divisor counts of perfect powers 35
2017 AIME I Problem 2: Equal remainders determine two divisors 35
2017 AIME I Problem 3: A periodic sequence of triangular-number digits 35
2017 AIME I Problem 5: One rational number in two bases 35
2017 AIME I Problem 9: A concatenation-type recurrence modulo99 35
2017 AIME I Problem 12: Product-free subsets 35
2017 AIME I Problem 13: The last gap between consecutive cubes 36
2017 AIME I Problem 14: Nested logarithms become a monotone power equation 36
2017 AIME II Problem 4: Ternary digit avoidance 36
2017 AIME II Problem 6: A square squeezed between nearby shifted squares 36
2017 AIME II Problem 8: Integrality of a truncated exponential series 36
2018 AIME I Problem 2: A digit permutation in three bases 36
2018 AIME I Problem 6: Two disjoint root-of-unity families 36
2018 AIME I Problem 11: A two-digit ending requires prime-square orders 36
2018 AIME I Problem 12: Subset sums modulo three 36
2018 AIME II Problem 2: A recurring triple proves periodicity 37
2018 AIME II Problem 3: A square and a cube in the same base 37

2018 AIME II Problem 7: Rational lengths among equal-area parallel slices . . . 37
2019 AIME I Problem 1: A sum of strings of nines . . . 37
2019 AIME I Problem 7: Prime exponents hidden in gcd and lcm logarithms . . . 37
2019 AIME I Problem 9: Consecutive integers with seven divisors in total . . . 37
2019 AIME I Problem 11: A circle tangent to all three excircles . . . 37
2019 AIME I Problem 14: An eighth power forces order sixteen . . . 37
2019 AIME II Problem 4: A square product of four die rolls . . . 38
2019 AIME II Problem 9: Multiples of twenty having twenty divisors . . . 38
2019 AIME II Problem 10: A tangent sign itinerary fixes every binary digit . . . 38
2019 AIME II Problem 12: Divisibility chains with prescribed sum . . . 38
2019 AIME II Problem 14: A three-denomination Frobenius problem . . . 38
2020 AIME I Problem 3: A shared set of digits in bases eleven and eight . . . 38
2020 AIME I Problem 4: A truncated prefix must divide the suffix . . . 38
2020 AIME I Problem 9: A random chain of divisors . . . 38
2020 AIME I Problem 10: A missing prime power constrains exponential divisibility . . . 39
2020 AIME I Problem 12: Prime-power divisibility after selecting an exponent class . . . 39
2020 AIME II Problem 1: Square divisors of a prime-power product . . . 39
2020 AIME II Problem 5: The smallest integer with nested digit-sum restrictions . . . 39
2020 AIME II Problem 10: Clear denominators before reducing a cube sum . . . 39
2021 AIME I Problem 3: Differences of powers of two . . . 39
2021 AIME I Problem 5: Integer arithmetic progressions satisfying a square identity . . . 39
2021 AIME I Problem 10: A rational recurrence reduces at its first shared prime . . . 39
2021 AIME I Problem 14: A universal divisor-sum congruence without advanced prime distribution . . . 40
2021 AIME II Problem 3: Cyclic products modulo three . . . 40
2021 AIME II Problem 9: Counting exponent pairs by valuations . . . 40
2021 AIME II Problem 13: Lift an exponential fixed point through powers of five . . . 40
2021 AIME II Problem 15: Walking to the next square with two step sizes . . . 40
2022 AIME I Problem 2: A cyclic digit shift in two bases . . . 40
2022 AIME I Problem 7: A minimal positive ratio of digit products . . . 40
2022 AIME I Problem 13: Distinct reduced numerators . . . 40
2022 AIME I Problem 14: Perimeter-bisecting cleavers determine a120-degree angle . . . 41
2022 AIME II Problem 1: A population ratio constrains the total size . . . 41
2022 AIME II Problem 5: Prime-difference triangles . . . 41
2022 AIME II Problem 8: Identifiability from three floor values . . . 41
2022 AIME II Problem 13: A coefficient reduced to stars and bars . . . 41
2022 AIME II Problem 14: Complete coverage by a finite stamp collection . . . 41
2023 AIME I Problem 4: Sum all divisors with prescribed exponent parity . . . 41
2023 AIME I Problem 7: Pairwise-distinct remainders . . . 41
2023 AIME I Problem 10: Bound the floor error before summing residues . . . 42
2023 AIME I Problem 15: A bounded Gaussian-integer search with all sign cases . . . 42
2023 AIME II Problem 2: Enumerate the four-digit octal palindromes . . . 42
2023 AIME II Problem 5: The cancellation divisor determines a rational number . . . 42
2023 AIME II Problem 10: Balanced proper colorings of a ladder . . . 42
2023 AIME II Problem 13: Prove the integrality class before counting last digits . . . 42
2023 AIME II Problem 15: A finite carry state detects repeated modular inverses . . . 42
2024 AIME I Problem 13: Lift the four roots of minus one . . . 42
2024 AIME II Problem 2: A mode and a missing median constrain an integer partition . . . 43
2024 AIME II Problem 6: Encoding a set by maxima of subsets . . . 43

2024 AIME II Problem 7: Sum digit replacements before solving the digits . . . 43
2024 AIME II Problem 13: Evaluate a root product at two conjugate constants . . . 43
2024 AIME II Problem 14: Digit-square conditions count CRT idempotents . . . 43
2025 AIME I Problem 1: Reduce a base-divisibility condition to one fixed integer . . . 43
2025 AIME I Problem 4: Count a union of two primitive lattice lines . . . 43
2025 AIME I Problem 5: Digit permutations divisible by twenty-two . . . 43
2025 AIME I Problem 15: Cubic residues and three-variable counting . . . 44
2025 AIME II Problem 2: Evaluate a polynomial at the modulus root . . . 44
2025 AIME II Problem 7: The least common multiple of a random divisor subset . . . 44
2025 AIME II Problem 8: When greedy coin change succeeds . . . 44
2025 AIME II Problem 13: Track reduction before simplifying a rational recurrence . . . 44
2026 AIME I Problem 4: Products of two unequal factors . . . 44
2026 AIME I Problem 6: A logarithmic quadratic determines a product . . . 44
2026 AIME I Problem 8: Divisor residues determined by exponent parity . . . 44
2026 AIME I Problem 13: Binomial residue sums over a finite field . . . 45
2026 AIME II Problem 1: Arithmetic progressions indexed by common divisors . . . 45
2026 AIME II Problem 3: An indented pentagon gives one residue class . . . 45
2026 AIME II Problem 4: A numeral equal to its value in a smaller base . . . 45
2026 AIME II Problem 5: Equal hypergeometric probabilities . . . 45
2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle . . . 45
2026 AIME II Problem 9: A divisor-generating decimal with controlled carries . . . 45
2026 AIME II Problem 10: An angle bisector and tangent secants . . . 45
2026 AIME II Problem 15: A Fano-plane residue count . . . 46

AIME Number Theory Problem Collection

New mathematical restatements of the standard 1983–2026 archive. Contest credit: MAA American Mathematics Competitions. Linked sources supply official wording and diagrams.

1983 AIME Problem 6: Powers one away from seven

Find the remainder of 6^83+8^83 upon division by 49.

Published problem and solutions · MAA AMC.

1983 AIME Problem 8: A prime divisor of a central binomial coefficient

Find the largest two-digit prime dividing C(200,100).

Published problem and solutions · MAA AMC.

1983 AIME Problem 12: A diameter with reversed chord digits

A circle has a two-digit integer diameter AB. A perpendicular chord CD has length obtained by reversing those digits. Their intersection H has positive rational distance from the center O. Find AB.

Published problem and solutions · MAA AMC.

1984 AIME Problem 2: The smallest zero-eight multiple

Let n be the smallest positive multiple of 15 whose decimal digits all belong to$\{0,8\}$. Find n/15.

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1984 AIME Problem 14: The last even number without two odd composites

Find the largest even integer that cannot be written as the sum of two odd composite positive integers.

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1985 AIME Problem 3: An integer complex cube

Positive integers a,b,c satisfy $c=(a+bi)^3-107i$, where $i^2=-1$. Find c.

Published problem and solutions · MAA AMC.

1985 AIME Problem 7: Two perfect powers nineteen apart

Positive integers a,b,c,d satisfy $a^5=b^4$, $c^3=d^2$, and c-a=19. Find d-b.

Published problem and solutions · MAA AMC.

1985 AIME Problem 13: The largest gcd of adjacent quadratic values

Let $a_n=100+n^2$ for positive integers n. Find the largest possible $\gcd(a_n,a_{n+1})$.

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1986 AIME Problem 5: A variable divisor and a constant remainder

Find the largest positive integer n such that n+10 divides n^3+100.

Published problem and solutions · MAA AMC.

1986 AIME Problem 6: A page counted twice

The page numbers 1 through n were summed, but one page number was added twice. The resulting total was 1986. Find the repeated page.

Published problem and solutions · MAA AMC.

1986 AIME Problem 7: Ranking subsets by ternary value

Arrange all positive sums of distinct nonnegative powers of 3 in increasing order. Find the hundredth term.

Published problem and solutions · MAA AMC.

1986 AIME Problem 8: Logarithms of proper divisors

Let S be the sum of the base 10 logarithms of all positive proper divisors of 1,000,000. Find the integer nearest to S.

Published problem and solutions · MAA AMC.

1987 AIME Problem 1: Addition without carrying

Count ordered nonnegative integer pairs (m,n) summing to 1492 with no decimal carry.

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1987 AIME Problem 3: Numbers equal to their proper-divisor product

For this problem, proper divisors exclude both 1 and the number itself. Call an integer n>1 nice when the product of its distinct proper divisors equals n. Find the sum of the first ten nice integers.

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1987 AIME Problem 5: A factored equation in two integer squares

Integers x,y satisfy $y^2+3x^2y^2=30x^2+517$. Find $3x^2y^2$.

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1987 AIME Problem 7: LCM triples

Count positive ordered triples (a,b,c) with lcm(a,b)=1000 and lcm(a,c)=lcm(b,c)=2000.

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1987 AIME Problem 8: Exactly one integer in a moving interval

Find the largest positive integer n for which exactly one integer k satisfies $8/15<n/(n+k)<7/13$.

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1987 AIME Problem 11: The longest consecutive-integer representation

Find the largest k such that 3^{11} is a sum of k consecutive positive integers.

Published problem and solutions · MAA AMC.

1987 AIME Problem 12: An integer just above a cube

Let m be the smallest integer with cube root n+r, where n is a positive integer and $0<r<1/1000$. Find n.

Published problem and solutions · MAA AMC.

1988 AIME Problem 2: A digit-sum iteration

Let T(k) be the square of the sum of the decimal digits of the positive integer k. Define $f_1(k)=T(k)$ and $f_n(k)=T(f_{n-1}(k))$. Find $f_{1988}(11)$.

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1988 AIME Problem 5: Divisor probability

A divisor of 10^99 is chosen uniformly. Find numerator plus denominator for its probability of being divisible by 10^88.

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1988 AIME Problem 8: A functional equation hiding the least common multiple

On positive integer pairs, f satisfies f(x,x)=x, f(x,y)=f(y,x), and (x+y)f(x,y)=y f(x,x+y). Find f(14,52).

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1988 AIME Problem 9: The smallest cube ending in eight-eight-eight

Find the least positive integer n whose cube has final three decimal digits 888.

Published problem and solutions · MAA AMC.

1989 AIME Problem 3: A repeated decimal digit forced by divisibility

For a positive integer n and a decimal digit d, n/810=0.d25d25d25…. Find n.

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1989 AIME Problem 4: One middle term, two perfect powers

Five consecutive positive integers are a<b<c<d<e. The sum b+c+d is a square, and a+b+c+d+e is a cube. Find the smallest possible c.

Published problem and solutions · MAA AMC.

1989 AIME Problem 9: Recovering a fifth root with congruences

A positive integer n satisfies $133^5+110^5+84^5+27^5=n^5$. Find n.

Published problem and solutions · MAA AMC.

1989 AIME Problem 14: Real integers written in a complex base

Let $\beta=-3+i$. Consider four-digit expressions $a_3\beta^3+a_2\beta^2+a_1\beta+a_0$ with decimal digits $0\le a_j\le 9$ and $a_3\neq 0$. Find the sum of all distinct real integers represented.

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1990 AIME Problem 1: Ranking nonsquares and noncubes

List positive integers that are neither perfect squares nor perfect cubes. Find the five-hundredth.

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1990 AIME Problem 3: A largest regular-polygon side count

Regular polygons with r and s sides satisfy r≥s≥3, and each interior angle of the r-gon is 59/58 times each interior angle of the s-gon. Find the largest possible s.

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1990 AIME Problem 5: The smallest multiple with seventy-five divisors

Let n be the smallest positive multiple of 75 having exactly 75 positive divisors. Find n/75.

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1990 AIME Problem 10: Products of two root-of-unity groups

Let $A = \{z : z^{18} = 1\}$ and $B = \{w : w^{48} = 1\}$. How many distinct complex numbers occur as zw with $z \in A$ and $w \in B$?

Published problem and solutions · MAA AMC.

1990 AIME Problem 11: A factorial as a shorter consecutive product

Find the largest positive integer n for which n! is the product of n−3 consecutive positive integers.

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1990 AIME Problem 13: Leading nines in powers

Among 9^0,…,9^4000, count those beginning with decimal digit 9, given that 9^4000 has 3817 digits and begins with 9.

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1991 AIME Problem 1: Symmetric sums with an integer restriction

Positive integers x,y satisfy xy+x+y=71 and x^2y+xy^2=880. Find x^2+y^2.

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1991 AIME Problem 5: Coprime factor allocations

How many reduced fractions strictly between 0 and 1 have numerator times denominator equal to 20!?

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1991 AIME Problem 6: Recovering a number from a floor sum

A real r satisfies $\Sigma_{k=19}{}^{91}$ floor(r+k/100)=546. Find floor(100r).

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1991 AIME Problem 8: Integer roots via a shifted factorization

For how many real values a does x^2+ax+6a=0 have two integer roots, counted with multiplicity?

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1991 AIME Problem 13: Sock counts from a probability

At most 1991 socks, red or blue, are in a drawer. Two drawn without replacement have the same color with probability 1/2. Maximize the number of red socks.

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1991 AIME Problem 15: An attainable vector minimum that is integral

For each positive integer n, let S_n be the minimum of $\Sigma_{k=1}{}^{n}\sqrt{((2k-1)^2+a_k{}^2)}$ over positive real a_k with Σa_k=17. Find the unique n for which S_n is an integer.

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1992 AIME Problem 1: Pairing reduced fractions around five

Find the sum of all positive rational numbers less than 10 whose reduced denominator is 30.

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1992 AIME Problem 5: Numerators of repeating decimals

For all numbers 0.abcabc… strictly between 0 and 1, count distinct numerators after reducing the fraction.

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1992 AIME Problem 6: Consecutive integers without carries

Count consecutive pairs whose members are both in [1000,2000] and whose decimal addition produces no carry.

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1992 AIME Problem 11: The order of two successive reflections

Reflect a line through the origin first in the line making angle $\pi/70$ with the positive x-axis, then in the line making angle $\pi/54$. Starting with y=(19/92)x, find the least positive number of repetitions that restores the original line.

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1992 AIME Problem 15: Missing factorial trailing-zero counts

A positive integer is attainable if it is the exact number of trailing zeros of some factorial. How many positive integers below 1992 are unattainable?

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1993 AIME Problem 4: Equal outer and inner sums

Count integer quadruples $0<a<b<c<d<500$ satisfying $a+d=b+c$ and $bc-ad=93$.

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1993 AIME Problem 6: One integer as three consecutive sums

Find the smallest positive integer expressible as the sum of nine consecutive integers, ten consecutive integers, and eleven consecutive integers.

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1993 AIME Problem 9: Colliding labels in a circular walk

On a circle of 2000 marked points, label one point 1. From label $k-1$ move k points clockwise to place label k, continuing through 1993. What is the smallest label sharing the point labeled 1993?

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1994 AIME Problem 1: Indexing squares one above multiples of three

The increasing sequence 3,15,24,48,… consists of positive multiples of 3 that are one less than a square. Find the remainder of its 1994 th term on division by 1000.

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1994 AIME Problem 4: A sum of binary logarithm floors

Find the positive integer n satisfying $\Sigma_{k=1}{}^{n}$ floor($\log_2$k)=1994.

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1994 AIME Problem 5: A weighted sum over digit strings

For each positive integer n, multiply its nonzero decimal digits. Let S be the sum of these products for n=1,…,999. Find the largest prime divisor of S.

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1994 AIME Problem 7: Lines with only integral circle intersections

Count real pairs (a,b) for which ax+by=1 and $x^2+y^2=50$ have at least one common point and every common point has integral coordinates.

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1994 AIME Problem 10: A prime-cube projection in an integer right triangle

Triangle ABC is right at C and has integer side lengths. The altitude from C meets AB at D with $BD=29^3$. If cosB=m/n in lowest terms, find m+n.

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1994 AIME Problem 11: Distinct heights of a brick tower

Build a tower of 94 bricks, each contributing height 4,10 or 19. Count distinct possible total heights.

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1994 AIME Problem 12: The finest square grid within a fence budget

A fenced 24 by 52 meter rectangle is partitioned completely into congruent axis-parallel square plots using at most 1994 meters of additional internal fence. Find the largest possible number of plots.

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1995 AIME Problem 2: A product of roots without solving them

Find the last three decimal digits of the product of all positive roots of $\sqrt{1995} \cdot x\hat{}(\log_{1995}x)=x^2$.

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1995 AIME Problem 6: Divisors below a square root

Let n=2^31 · 3^19. Count divisors of n^2 that are smaller than n but do not divide n.

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1995 AIME Problem 8: Two simultaneous integer ratios

Count positive integer pairs $y < x \leq 100$ for which x/y and $(x+1)/(y+1)$ are integers.

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1995 AIME Problem 10: A composite term in five consecutive residue steps

Find the largest positive integer that cannot be expressed as a positive multiple of 42 plus a positive composite integer.

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1995 AIME Problem 11: A self-similar cut prism

An integer-sided rectangular prism has sorted sides $a \leq b \leq c$ with $b = 1995$. A face-parallel cut produces a smaller similar prism. Count possible triples (a, b, c).

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1995 AIME Problem 13: Grouping a sum by rounded fourth roots

Let f(n) be the integer nearest to the positive fourth root of n. Find $\Sigma_{k=1}{}^{1995} 1/f(k)$.

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1996 AIME Problem 2: Even logarithm floors

Count positive integers $n < 1000$ for which $\lfloor \log_2 n \rfloor$ is positive and even.

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1996 AIME Problem 8: Integer pairs of prescribed harmonic mean

Count positive integer pairs x<y whose harmonic mean is 6^20.

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1996 AIME Problem 10: An integer angle from a tangent identity

Find the smallest positive integer x satisfying tan(19x°)=(cos 96°+sin 96°)/(cos 96°−sin 96°).

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1996 AIME Problem 14: Cubes crossed by a space diagonal

A box of side lengths 150,324,375 is partitioned into unit cubes. Count cubes whose interiors meet a long diagonal.

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1997 AIME Problem 1: Differences of squares

Count integers 1 through 1000 that are differences of squares of nonnegative integers.

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1997 AIME Problem 3: Concatenation mistaken for multiplication

A two-digit integer x written immediately before a three-digit integer y forms a five-digit number equal to nine times xy. Find x+y.

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1997 AIME Problem 5: Decimal numbers closest to two sevenths

Count four-place decimals 0.abcd for which 2/7 is the closest fraction whose numerator is 1 or 2 and whose denominator is an integer.

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1997 AIME Problem 6: An equilateral attachment and another regular polygon

An equilateral triangle A_1A_2B is attached externally to a regular n-gon $A_1A_2\ldots A_n$. Find the largest n for which A_1,A_n,B can be three consecutive vertices of a regular polygon.

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1998 AIME Problem 1: Counting possible LCM partners

How many positive integers k satisfy lcm(6^6,8^8,k)=12^12?

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1998 AIME Problem 5: A four-term cycle of signed triangular numbers

For positive k let A_k=k(k−1)cos(k(k−1)π/2)/2. Find $|A_{19}+A_{20}+\cdots+A_{98}|$.

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1998 AIME Problem 8: The longest subtractive recurrence

A sequence begins 1000,x and each later term is the term two places earlier minus the preceding term. Stop when the first negative term appears, including it. Which positive integer x makes the sequence longest?

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1998 AIME Problem 14: A sharp bound for a box volume ratio

Positive integers m≤n≤p satisfy 2 mnp=(m+2)(n+2)(p+2). Find the largest possible p.

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1999 AIME Problem 1: Five primes in arithmetic progression

Find the smallest prime that can be the fifth term of a strictly increasing arithmetic progression whose first four terms are also prime.

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1999 AIME Problem 3: A quadratic square condition with four solutions

Find the sum of all positive integers n for which $n^2-19n+99$ is a perfect square.

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1999 AIME Problem 5: Possible digit-sum changes

Let S(x) be the decimal digit sum and T(x)=|S(x+2)−S(x)| for positive integers x. Count distinct T-values at most 1999.

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1999 AIME Problem 7: Four-position divisor switches

Switches labeled $2^x3^y5^z$ for $0 \le x, y, z \le 9$ cycle $A \to B \to C \to D \to A$. Processing every label once advances all switches whose labels divide it. Starting at A, how many finish at A?

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1999 AIME Problem 13: Distinct tournament win totals

Forty teams play each pair once with equally likely independent winners. If the probability that all win totals differ is m/n in lowest terms, find $\log_2 n$.

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2000 AIME I Problem 1: Zero-free factors of powers of ten

Find the smallest positive integer n such that, whenever positive integers a, b satisfy $ab = 10^n$, at least one of a, b has a zero in its decimal expansion.

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2000 AIME I Problem 2: A reflected pentagon

Let $A = (u, v)$, where u, v are integers with $0 < v < u$. Obtain B, C, D, E successively by reflecting A across $y = x$, B across the y-axis, C across the x-axis, and D across the y-axis. If pentagon $ABCDE$ has area 451, find $u + v$.

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2000 AIME I Problem 5: Two urns with a fixed total

Two urns each contain both black and white balls, with 25 balls in total. Independent uniform draws, one per urn, are both black with probability 27/50. Write the probability of two whites as reduced m/n and find $m + n$.

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2000 AIME I Problem 6: A gap between means

How many integer pairs (x, y) satisfy $0 < x < y < 10^6$ and $\frac{x+y}{2} - \sqrt{xy} = 2$?

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2000 AIME I Problem 11: Independent prime exponents

Sum a/b over ordered pairs of coprime positive divisors of 1000. If the sum is S, find $\lfloor S/10 \rfloor$.

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2000 AIME II Problem 2: Lattice points on a difference of squares

Count all ordered integer pairs (x, y) satisfying $x^2 - y^2 = 2000^2$.

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2000 AIME II Problem 4: Prescribed odd and even divisor counts

Find the least positive integer having exactly six positive odd divisors and twelve positive even divisors.

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2000 AIME II Problem 11: Integral equal legs restrict base directions

An isosceles trapezoid $ABCD$ has integer-coordinate vertices, $A = (20, 100)$, $D = (21, 107)$, no horizontal or vertical side, and only $AB \parallel CD$. The sum of the absolute values of all possible slopes of AB is m/n in lowest terms. Find $m + n$.

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2000 AIME II Problem 14: Borrowing in factorial base

For $N = 16! - 32! + 48! - \cdots - 1984! + 2000!$, let $N = \sum_{i=1}^{j} f_i i!$ be its factorial-base expansion, where $0 \le f_i \le i$ and $f_j > 0$. Find $\sum_{i=1}^{j} (-1)^{i+1} f_i$.

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2001 AIME I Problem 1: Two digits dividing their number

Find the sum of all positive two-digit integers divisible by both of their digits.

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2001 AIME I Problem 8: Largest number with doubled reinterpreted digits

Call $N > 0$ admissible if reading its base-7 digits as a decimal numeral gives $2N$. Find the largest admissible integer.

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2001 AIME I Problem 11: Two numberings of a rectangular array

In a $5 \times N$ array choose one point per row. Row-major labels x_i and column-major labels y_i obey $x_1 = y_2, x_2 = y_1, x_3 = y_4, x_4 = y_5, x_5 = y_3$. Find the smallest N.

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2001 AIME II Problem 1: Digit adjacency as a directed graph

Find the leading three digits of the largest positive integer whose every adjacent two-digit block is a perfect square.

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2001 AIME II Problem 5: Triangle-free ordered sets

Every ten-element subset of $\{4, 5, \ldots, n\}$ must contain three side lengths of a nondegenerate triangle. Find the largest such n.

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2001 AIME II Problem 10: Counting exponent gaps

How many different numbers $10^j - 10^i$, with $0 \le i < j \le 99$, are divisible by 1001?

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2001 AIME II Problem 14: Unit roots of a sparse polynomial

List the unit-modulus roots of $z^{28} - z^8 - 1 = 0$ by arguments $0 \le \theta_1 < \cdots < \theta_{2n} < 360^\circ$. Find $\theta_2 + \theta_4 + \cdots + \theta_{2n}$.

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2002 AIME I Problem 3: Reversed two-digit ages

One person is currently 25, and another is older. In a positive integer number n of years their ages are two-digit reversals. Count the possible ordered pairs consisting of the older person's present age and n.

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2002 AIME I Problem 4: A telescope followed by a prime-square factorization

For $a_k = 1/(k^2 + k)$, positive integers $m < n$ satisfy $\sum_{k=m}^{n-1} a_k = 1/29$. Find $m + n$.

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2002 AIME I Problem 7: Three fractional digits of an enormous power

Find the first three digits after the decimal point of $(10^{2002} + 1)^{10/7}$.

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2002 AIME I Problem 8: The first repeated Fibonacci-type endpoint

Consider nondecreasing positive integer sequences satisfying $a_n = a_{n-1} + a_{n-2}$ for $n > 2$. Find the smallest value of a_9 attained by at least two such sequences.

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2002 AIME I Problem 9: Three arithmetic progressions partition the integers

Three progressions start at $1, 2, 3$ and have positive integer steps h, t, u. They partition the positive integers exactly once. Sum $100h + 10t + u$ over all valid triples.

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2002 AIME I Problem 11: Unfolding a reflected ray in a cube

A ray starts at a vertex of a cube of side 12. Its first reflection is on the opposite face at offsets 7 and 5 from the two adjacent edges through the corresponding vertex. It reflects specularly until reaching another cube vertex. If the path length is $m\sqrt{n}$ with square-free n, find $m + n$.

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2002 AIME I Problem 14: Integer means after deletion

A set of distinct positive integers contains 1, has maximum 2002, and has integral mean after deleting any single member. Find its largest possible size.

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2002 AIME II Problem 1: Differences of reversed three-digit integers

Let x, y be three-digit positive integers whose decimal digits are reversals of one another. How many distinct values of $|x - y|$ occur?

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2002 AIME II Problem 3: An integral geometric progression with a square gap

Positive integers $a < b < c$ form a geometric progression and satisfy $\log_6 a + \log_6 b + \log_6 c = 6$. If $b - a$ is a perfect square, find $a + b + c$.

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2002 AIME II Problem 5: When a power fails to divide another

Find the sum of all integers $a = 2^n 3^m$, with $n, m \geq 0$, for which a^6 does not divide 6^a.

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2002 AIME II Problem 7: A square sum divisible by 200

Find the smallest positive integer k for which $1^2 + 2^2 + \cdots + k^2$ is divisible by 200.

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2002 AIME II Problem 8: The first missing integer quotient

Find the least positive integer k never equal to $\lfloor 2002/n \rfloor$ for an integer $n \neq 0$.

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2003 AIME I Problem 1: The largest factorial factor

Write $((3!)!)/3! = k\,n!$ with positive integers k,n and n as large as possible. Find $k + n$.

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2003 AIME I Problem 7: Integral distances in an isosceles triangle

Points A, B, C are collinear in that order with $AB = 9, BC = 21$. A noncollinear point D satisfies $AD = CD$, and both AD and BD are integers. Find the sum of all possible perimeters of triangle ACD.

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2003 AIME I Problem 8: An arithmetic triple followed by a geometric triple

Four increasing positive integers have their first three in arithmetic progression and their last three in geometric progression. The last exceeds the first by 30. Find their sum.

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2003 AIME I Problem 14: The smallest denominator containing a decimal block

For reduced positive fractions $m/n < 1$, find the smallest denominator n whose decimal expansion contains the consecutive digits 251 in that order.

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2003 AIME II Problem 1: A product equal to six times a sum

Three positive integers have product N equal to six times their sum. One integer is the sum of the other two. Find the sum of all possible N.

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2003 AIME II Problem 2: Largest distinct-digit multiple of eight

Let N be the largest positive multiple of 8 with no repeated decimal digit. Find N mod 1000.

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2003 AIME II Problem 10: A nonsquare radical sum with fixed difference

Two positive integers differ by 60, and the sum of their square roots is the square root of a nonsquare integer. Find their largest possible sum.

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2003 AIME II Problem 12: Minimum occupancy with percentage constraints

Twenty-seven candidates receive integer vote counts totaling s. For each candidate, its vote count exceeds its exact percentage of all votes by at least one. Find the smallest s.

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2004 AIME I Problem 1: Four descending consecutive digits

A four-digit positive integer has consecutive digits in strictly decreasing order. Find the sum of all possible remainders modulo 37.

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2004 AIME I Problem 5: Strict daily rates with a larger aggregate score

A contestant attempts 500 points over two days and scores a positive integer each day. Its first-day success rate is below 160/300, its second-day rate below 140/200, and it does not attempt 300 points on day one. Find m+n for the largest possible aggregate rate m/n in lowest terms.

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2004 AIME I Problem 8: Regular star polygons

Count similarity classes of connected regular 1000-vertex star polygons with equal edge lengths and equal counterclockwise turning angles less than 180°, requiring a genuine edge crossing.

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2004 AIME I Problem 15: Reverse dynamics and forbidden operation runs

Iterate $f(1) = 1$, $f(x) = x/10$ when $10 \mid x$, and $f(x) = x + 1$ otherwise. Let $d(x)$ be the first index with value 1, starting at index one. If m integers have $d(x) = 20$, find the sum of distinct prime factors of m.

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2004 AIME II Problem 3: Hidden cubes

A rectangular block made of unit cubes has exactly 231 cubes invisible when three mutually adjacent faces are viewed. Find the least possible total number of cubes.

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2004 AIME II Problem 6: Integer shares in a three-stage distribution

Three monkeys successively take portions of a banana pile. They keep respectively $3/4, 1/4, 1/12$ of their portions and split each remainder equally between the other two. Every transfer is integral, the last monkey empties the pile, and final holdings have ratio $3 : 2 : 1$. Find the smallest initial pile.

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2004 AIME II Problem 8: Divisor counts as exponent allocations

How many divisors of 2004^{2004} have exactly 2004 positive divisors?

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2004 AIME II Problem 10: Two binary ones modulo nine

Choose uniformly an integer below 2^{40} with exactly two binary ones. Find numerator plus denominator of the probability that it is divisible by 9.

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2004 AIME II Problem 14: Distinct digit lengths of a sum

A string of n sevens is split into decimal integers by inserting plus signs. How many values of n can give a total of 7000?

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2005 AIME I Problem 2: Arithmetic progressions reaching 2005

For each positive integer k, consider $1, 1+k, 1+2k, \ldots$. How many values of k make 2005 a term?

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2005 AIME I Problem 3: Exactly three small proper divisors

How many positive integers have exactly three positive proper divisors, all less than 50?

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2005 AIME I Problem 4: A rectangular formation five above a square

A band has five more members than a square formation can hold, and exactly fills a rectangle with seven more rows than columns. Find its greatest possible size.

Published problem and solutions · MAA AMC.

2005 AIME I Problem 12: Parity changes only at squares

Let $S(n) = \sum_{k=1}^{n} \tau(k)$, where τ counts positive divisors. Among $1 \le n \le 2005$, let a and b count odd and even values of $S(n)$. Find $|a-b|$.

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2005 AIME II Problem 4: Union of divisor sets

Count positive integers dividing at least one of $10^{10}, 15^7, 18^{11}$.

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2005 AIME II Problem 5: Reciprocal logarithms force powers

Count ordered integer pairs (a, b) with $2 \le a, b \le 2005$ satisfying $\log_a b + 6 \log_b a = 5$.

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2005 AIME II Problem 6: Fixed points of a shuffle

A stack labeled $1, \ldots, 2n$ top to bottom is cut into two equal halves. Restack by alternately placing the next card from the lower half and then the upper half on top. At least one card from each half stays fixed, including card 131. Find $2n$.

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2005 AIME II Problem 9: A swapped sine-cosine power identity

For how many positive integers $n \leq 1000$ does $(\sin t + i \cos t)^n = \sin(nt) + i \cos(nt)$ hold for every real t?

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2005 AIME II Problem 13: Integer polynomial values constrain integer roots

An integer-coefficient polynomial satisfies $P(17) = 10$, $P(24) = 17$, and $P(n) = n + 3$ for two distinct integers n_1, n_2. Find $n_1 n_2$.

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2006 AIME I Problem 3: Deleting the leading digit

Find the smallest positive integer whose value becomes one twenty-ninth as large when its leading decimal digit is removed.

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2006 AIME I Problem 4: Valuation of a product of factorials

Let z be the number of trailing decimal zeros of $\prod_{n=1}^{100} n!$. Find z modulo 1000.

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2006 AIME I Problem 9: Integral geometric progressions and logarithms

A geometric progression begins with positive integer a and has positive integer ratio r. If $\sum_{k=0}^{11} \log_8(ar^k) = 2006$, how many pairs (a, r) exist?

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2006 AIME I Problem 13: Valuation layers

For even x, let $g(x)$ be the largest power of two dividing x. Set $S_n = \sum_{k=1}^{2^{n-1}} g(2k)$. Find the greatest $n < 1000$ for which S_n is a square.

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2006 AIME II Problem 3: Three-adic valuation of an odd factorial

Find the greatest exponent k for which 3^k divides $1 \cdot 3 \cdot 5 \cdots 199$.

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2006 AIME II Problem 7: Complementary decimal numbers

Count ordered positive integer pairs (a, b) with $a + b = 1000$ and no zero digit in either number.

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2006 AIME II Problem 13: Five representations by consecutive odd integers

How many integers $N < 1000$ have exactly five representations as a sum of one or more consecutive positive odd integers, counting distinct sequence lengths?

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2006 AIME II Problem 14: Reciprocal digit frequency

Let S_n be the sum of reciprocals of all nonzero decimal digits appearing in $1, \ldots, 10^n$. Find the least positive n for which S_n is integral.

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2007 AIME I Problem 1: Square multiples of twenty-four

How many positive perfect squares below 10^6 are divisible by 24?

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2007 AIME I Problem 4: The next planetary alignment

Three coplanar planets travel uniformly in circular orbits in the same direction, with periods $60, 84, 140$ years. They and the central star are initially collinear. After how many years will this first recur?

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2007 AIME I Problem 5: A periodic rounding map

Convert an integer Fahrenheit temperature in $[32, 1000]$ to Celsius using $C = 5(F-32)/9$, round to the nearest integer, convert back, and round again. Count temperatures recovered exactly.

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2007 AIME I Problem 7: A floor-ceiling logarithm indicator

Find $\sum_{k=1}^{1000} k(\lceil \log_{\sqrt{2}} k \rceil - \lfloor \log_{\sqrt{2}} k \rfloor)$ modulo 1000.

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2007 AIME I Problem 11: Rounding square roots in blocks

For each positive integer p, let $b(p)$ be its square root rounded to the nearest integer. Find $\sum_{p=1}^{2007} b(p)$ (mod 1000).

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2007 AIME II Problem 2: Divisibility becomes a composition

Count ordered positive triples (a, b, c) with $a \mid b$, $a \mid c$, and $a + b + c = 100$.

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2007 AIME II Problem 5: Grid cells below a rational line

Count unit grid squares whose interiors lie in the first quadrant and below $9x + 223y = 2007$.

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2007 AIME II Problem 7: Multiples inside a cube-root interval

Exactly 70 positive integers n satisfy both $\lfloor \sqrt[3]{n} \rfloor = k$ and $k \mid n$ for some integer k. Find the largest possible value of n/k among them.

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2007 AIME II Problem 12: The largest term controls a geometric sum

An increasing geometric sequence consists of integral powers of 3. It satisfies $\sum_{j=0}^{7} \log_3 x_j = 308$ and $56 \leq \log_3(\sum_{j=0}^{7} x_j) \leq 57$. Find $\log_3 x_{14}$.

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2007 AIME II Problem 13: Pascal propagation modulo three

Place eleven binary digits in the bottom row of a triangular array. Each upper entry is the sum of its two children. Count assignments making the top entry divisible by 3.

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2008 AIME I Problem 3: Integer speeds from two distance totals

Biking, jogging, and swimming speeds b, j, s are positive integers. A two-hour bike ride, three-hour jog, and four-hour swim cover 74 km; four hours biking, two jogging, and three swimming cover 91 km. Find $b^2 + j^2 + s^2$.

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2008 AIME I Problem 4: A shifted difference of squares

Positive integers x, y satisfy $x^2 + 84x + 2008 = y^2$. Find $x + y$.

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2008 AIME I Problem 6: Summing an arithmetic row repeatedly

The top row is $1, 3, \ldots, 99$. Each row below sums adjacent entries until one remains. Count all entries divisible by 67.

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2008 AIME I Problem 7: Occupied hundred-blocks

For $0 \le i \le 999$, let $S_i = \{100i, \ldots, 100i + 99\}$. How many of these sets contain no perfect square?

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2008 AIME II Problem 4: Balanced ternary support

Write 2008 as a sum of distinct nonnegative powers of 3, each with coefficient 1 or -1. Find the sum of the exponents that occur.

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2008 AIME II Problem 8: A trigonometric telescope and divisibility

Put $a = \pi/2008$. Find the smallest positive integer n for which $2\sum_{k=1}^{n} \cos(k^2 a)\sin(ka)$ is an integer.

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2008 AIME II Problem 15: Consecutive cubes and a second square

Find the largest integer n such that n^2 is the difference of consecutive integer cubes and $2n + 79$ is a perfect square.

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2009 AIME I Problem 1: Extremal three-digit geometric progressions

A three-digit integer is called geometric when its three distinct digits, in order, form a geometric progression. Find the largest such integer minus the smallest.

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2009 AIME I Problem 6: Integer values of a stepped power function

How many positive integers $N < 1000$ occur as $x^{\lfloor x \rfloor}$ for some real x where the expression is defined?

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2009 AIME I Problem 7: An exponential recurrence that telescopes

Let $a_1 = 1$ and $5^{a_{n+1} - a_n} - 1 = 1/(n + 2/3)$ for $n \ge 1$. Find the least integer $k > 1$ such that a_k is integral.

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2009 AIME I Problem 11: Integral triangle areas on a line

Choose distinct nonnegative lattice points P, Q on $41x + y = 2009$. Count triangles OPQ of positive integer area, where $O = (0, 0)$.

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2009 AIME I Problem 13: Infinite integrality forces a two-cycle

Every term of a sequence is a positive integer, and $a_{n+2} = (a_n + 2009)/(1 + a_{n+1})$ for $n \geq 1$. Find the smallest possible $a_1 + a_2$.

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2009 AIME I Problem 14: Minimizing a second moment with fixed first and fourth moments

A sequence of 350 entries, each in $\{1, 2, 3, 4\}$, satisfies $\sum a_i = 513$ and $\sum a_i^4 = 4745$. Find the smallest possible $\sum a_i^2$.

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2009 AIME II Problem 4: The smallest winning score

For some number of children, the child ranked k eats exactly $n + 2 - 2k$ grapes, a nonnegative amount. Their total is 2009. Find the smallest possible winning count n.

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2009 AIME II Problem 7: A sum with strictly increasing dyadic denominators

Write $\sum_{i=1}^{2009} (2i-1)!!/(2i)!!$ in lowest terms with denominator $2^a b$, where b is odd. Find $ab/10$.

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2009 AIME II Problem 9: An injection leaves two boundary slices

Let m, n count positive integer solutions to $4x + 3y + 2z = 2009$ and $4x + 3y + 2z = 2000$, respectively. Find $(m - n) \bmod 1000$.

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2009 AIME II Problem 11: Exactly fifty logarithmic neighbors

Positive integers $m \geq n$ have exactly fifty positive integers k satisfying $|\log m - \log k| < \log n$. Find the sum of all possible products mn.

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2010 AIME I Problem 1: One square divisor in a pair

Choose two distinct positive divisors of 2010^2 uniformly. Find numerator plus denominator of the probability that exactly one is a square.

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2010 AIME I Problem 2: A product of repeated nines

Find $\prod_{k=1}^{999} (10^k - 1)$ modulo 1000.

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2010 AIME I Problem 5: Equality in a difference-of-squares bound

Positive integers $a > b > c > d$ satisfy $a + b + c + d = 2010$ and $a^2 - b^2 + c^2 - d^2 = 2010$. How many distinct values can a take?

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2010 AIME I Problem 8: The smallest disk enclosing floor-defined squares

Let $R = \{(x, y) : \lfloor x \rfloor^2 + \lfloor y \rfloor^2 = 25\}$. The smallest closed disk containing R has radius $\sqrt{m}/n$ in reduced squarefree form. Find $m + n$.

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2010 AIME I Problem 10: Digits allowed to carry

Count representations $2010 = 1000a_3 + 100a_2 + 10a_1 + a_0$ with every integer a_i between 0 and 99.

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2010 AIME I Problem 12: A multiplicative two-color threshold

Find the least $m \geq 3$ such that every two-coloring of $\{3, \ldots, m\}$ has a monochromatic solution $ab = c$, allowing $a = b$.

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2010 AIME I Problem 14: A monotone sum of decimal digit counts

For positive integers n, define $f(n) = \sum_{k=1}^{100} \lfloor \log_{10}(kn) \rfloor$. Find the largest n with $f(n) \leq 300$.

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2010 AIME II Problem 1: The largest multiple with distinct even digits

Let N be the largest positive multiple of 36 whose decimal digits are distinct and even. Find N modulo 1000.

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2010 AIME II Problem 3: Valuation of all pair differences

Let $K = \prod_{1 \leq a < b \leq 20} (b - a)$. Find the largest exponent n for which $2^n \mid K$.

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2010 AIME II Problem 6: The first reducible quartic

Find the smallest positive integer n for which $x^4 - nx + 63$ factors into two nonconstant integer-coefficient polynomials.

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2010 AIME II Problem 10: Quadratics modulo interchange of roots

Count quadratic polynomials with integer coefficients and integer roots whose constant coefficient is 2010.

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2010 AIME II Problem 12: Two isosceles triangles with equal area and perimeter

Two noncongruent integer-sided isosceles triangles have equal areas and perimeters, and their base lengths are in ratio $8:7$. Find their least possible common perimeter.

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2011 AIME I Problem 5: Sliding sums force periodic residues

Label a regular nonagon by $1, \ldots, 9$, requiring every three consecutive labels to have sum divisible by 3. Identify rotations but not reflections. Count arrangements.

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2011 AIME I Problem 7: Splitting a power into equal smaller powers

For how many positive integers m can one power m^{x_0} equal a sum of exactly 2011 powers m^{x_i} with all exponents nonnegative?

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2011 AIME I Problem 10: Obtuse triangles in a regular polygon

Three vertices are chosen uniformly from a regular n-gon. Their triangle is obtuse with probability $93/125$. Find the sum of all possible n.

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2011 AIME I Problem 11: Summing a power-residue orbit without its order

Let R be the set of distinct residues of 2^n modulo 1000 for integers $n \geq 0$. Find the remainder modulo 1000 of the sum of the elements of R.

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2011 AIME I Problem 15: Bounding integer roots of a depressed cubic

The polynomial $x^3 - 2011x + m$ has three integer roots. Find the sum of their absolute values.

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2011 AIME II Problem 3: Integer angles in arithmetic progression

The interior angle measures of a convex 18-gon, in increasing order, form a strictly increasing arithmetic progression of integers. Find its smallest angle in degrees.

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2011 AIME II Problem 14: Permutations respecting residue partitions

Count permutations $a_1, \ldots, a_{30}$ of $1, \ldots, 30$ satisfying $a_{i+m} \equiv a_i \pmod{m}$ for $m = 2, 3, 5$ wherever defined. Give the residue modulo 1000.

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2011 AIME II Problem 15: Floor equality on three intervals

Let $P(x) = x^2 - 3x - 9$ and choose x uniformly on $[5, 15]$. The probability that $\lfloor \sqrt{P(x)} \rfloor = \sqrt{P(\lfloor x \rfloor)}$ has form $(\sqrt{a} + \sqrt{b} + \sqrt{c} - d)/e$. Find $a + b + c + d + e$.

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2012 AIME I Problem 1: Divisibility in both digit directions

Count three-digit integers abc with $a, c \neq 0$ such that both abc and cba are divisible by four.

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2012 AIME I Problem 4: A shared horse and the first integer meeting point

Two travelers start together along a road with mileposts at every positive integer distance. Their walking speeds are 4 and $5/2$ mph; their shared horse carries either at 6 mph. The slower walker rides first to post one and leaves the horse. Thereafter each rider passes the other traveler, leaves the horse at the next post, and walks on. At their first meeting at a milepost, the distance is n miles and elapsed time is t minutes. Find $n + t$.

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2012 AIME I Problem 5: Consecutive binary numbers with equal weight

Let B contain all thirteen-bit strings with eight ones and five zeros, allowing leading zeros. Count ordered pairs $(u, v) \in B^2$ with numeric difference $u - v = 1$.

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2012 AIME I Problem 6: A positive imaginary part selects a root-of-unity denominator

Complex numbers satisfy $z^{13} = w$ and $w^{11} = z$. Suppose $\operatorname{Im} z = \sin(m\pi/n)$ for coprime positive integers $m < n$. Determine n.

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2012 AIME I Problem 10: Order the squares with a prescribed three-digit suffix

For every perfect square whose last three decimal digits are 256, delete those digits, allowing the result zero from the square 256. Arrange the distinct resulting nonnegative integers in increasing order. Find the tenth one modulo 1000.

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2012 AIME I Problem 11: Reachability in a lattice semigroup

From $(0,0)$ allow jumps $(7,2),(2,7),(-5,-10),(-10,-5)$. Let M count reachable lattice points with $|x|+|y|\le 100$. Find M mod 1000.

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2012 AIME I Problem 15: Multiplication changes every circular separation

Reseat n people on a numbered circle by $k \mapsto ak \pmod n$. Require this to be a permutation and require every pair's circular separation to change even up to reversal. Count possible $1 < n < 1000$.

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2012 AIME II Problem 1: Count lattice points on a positive Diophantine segment

Count ordered positive integer pairs (m,n) satisfying $20m+12n=2012$.

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2012 AIME II Problem 12: Independent forbidden residues

A positive integer is p-safe if its distance from every multiple of p exceeds two. Count integers at most 10000 that are simultaneously 7-, 11-, and 13-safe.

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2013 AIME I Problem 2: Uniform digit sums modulo five

Count five-digit integers divisible by five whose first and last digits agree and whose digit sum is divisible by five.

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2013 AIME I Problem 8: A function domain imposes an integer divisor

For positive integers $m>1, n$, the real domain of $\arcsin(\log_m(nx))$ has length $1/2013$. Find the least possible $m+n$, reduced modulo 1000.

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2013 AIME I Problem 10: Conjugate integer roots and distinct cubics

Consider all cubics $x^3-ax^2+bx-65$ with nonzero integers a,b that have a root $r+si$ for nonzero integers r,s. For each distinct coefficient pair (a,b), take the sum of its three roots. Find the sum of these values over all admissible pairs.

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2013 AIME I Problem 11: Three remainder conditions after an lcm reduction

A positive integer N is divisible by $14, 15, 16$. There exist three distinct positive integers below 14 for which dividing N leaves remainder 3. Find the sum of the distinct prime divisors of the least such N.

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2013 AIME I Problem 15: Two arithmetic progressions with reordered residues

Count $0 \le A < B < C \le 99$ forming an arithmetic progression for which a prime p and residues $0 \le b < a < c < p$ satisfy $A \equiv a$, $B \equiv b$, $C \equiv c \pmod{p}$, with (b, a, c) also an arithmetic progression.

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2013 AIME II Problem 2: Nested logarithms impose a power-of-two divisor

For positive integers a, b, suppose $\log_2(\log_{2^a}(\log_{2^b}(2^{1000}))) = 0$. Sum the distinct possible values of $a+b$.

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2013 AIME II Problem 6: The first complete thousand-block between squares

Find the least positive integer N for which none of $1000N, 1000N + 1, \ldots, 1000N + 999$ is a perfect square.

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2013 AIME II Problem 7: Staff reductions force a unique feasible integer solution

Each clerk sorts 30 files per hour. Starting with x clerks, the same positive number t leaves after each of the first three hours. The remaining group finishes 1775 files at time 3 hours 10 minutes. How many files were sorted during the first 90 minutes?

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2013 AIME II Problem 14: Maximize a remainder by separating quotient ranges

For an integer $n > 1$, let $F(n)$ be the largest remainder obtained when n is divided by an integer k with $1 \le k \le n/2$. Find $\sum_{n=20}^{100} F(n)$ modulo 1000.

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2014 AIME I Problem 3: Reduced fractions with fixed numerator-plus-denominator

Count reduced fractions strictly between zero and one whose positive numerator and denominator sum to 1000.

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2014 AIME I Problem 6: Two parabolas with a shared axis

The parabolas $y = 3(x-h)^2 + j$ and $y = 2(x-h)^2 + k$ have vertical intercepts 2013 and 2014. Each has two positive integer horizontal intercepts. Find h.

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2014 AIME I Problem 8: A four-digit idempotent

A positive integer and its square end in the same four digits, whose thousands digit is nonzero. Find the first three digits of that four-digit ending.

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2014 AIME II Problem 4: Compatible repeating blocks

Digits a, b, c satisfy $0.\overline{ab} + 0.\overline{abc} = 33/37$. Find the three-digit integer $100a + 10b + c$.

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2014 AIME II Problem 15: Prime multiplication as binary carrying

For positive k, let $p(k)$ be the smallest prime not dividing k, and let $X(k)$ be the product of smaller primes, with empty product 1. Define $x_0 = 1$ and $x_{n+1} = x_n p(x_n)/X(x_n)$. Find the first $t > 0$ with $x_t = 2090$.

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2015 AIME I Problem 3: A prime concealed in a cube

Find the prime p such that $16p + 1$ is a positive perfect cube.

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2015 AIME I Problem 8: Digit sums determine the carries

Find the least positive integer n such that both n and $n + 864$ have digit sum 20.

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2015 AIME I Problem 11: Integral isosceles triangles with a fixed incenter distance

An isosceles triangle has $AB = AC$, positive integer side lengths, and incenter I satisfying $BI = 8$. Find its minimum perimeter.

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2015 AIME II Problem 1: Two percentage relations force a least common multiple

A positive integer N is 22 percent less than one integer and 16 percent greater than another. For the smallest such N, find its remainder modulo 1000.

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2015 AIME II Problem 3: A multiple with a prescribed digit sum

Find the least positive multiple of 17 whose decimal digits sum to 17.

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2015 AIME II Problem 6: Two possible cubics with positive integer roots

The polynomial $2x^3 - 2ax^2 + (a^2 - 81)x - c$ has three positive integer roots, counted with multiplicity. Knowing a still leaves exactly two possible values of c. Find their sum.

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2015 AIME II Problem 8: An integer inequality bounds a cubic quotient

Positive integers a, b satisfy $(ab+1)/(a+b) < 3/2$. If the largest possible value of $(a^3b^3+1)/(a^3+b^3)$ is p/q in lowest terms, find $p+q$.

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2015 AIME II Problem 13: Negative partial sums of successive sines

For $n \geq 1$, let $a_n = \sum_{k=1}^{n} \sin k$, with angles in radians. Find the index of the hundredth negative term.

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2016 AIME I Problem 5: Two arithmetic totals share a day count

A reader starts with n pages in t minutes, then increases both quantities by one each day. She reads on at least two days, finishing 374 pages in 319 minutes; n, t are positive integers. Find $n+t$.

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2016 AIME I Problem 7: Real values of a complex radical expression

Using principal square roots, count ordered integer pairs (a, b) for which $\sqrt{ab+2016}/(ab+100) - i\sqrt{|a+b|}/(ab+100)$ is defined and real.

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2016 AIME I Problem 10: Alternating geometric and arithmetic triples

A strictly increasing positive integer sequence has every triple $a_{2k-1}, a_{2k}, a_{2k+1}$ geometric and every triple $a_{2k}, a_{2k+1}, a_{2k+2}$ arithmetic. Given $a_{13} = 2016$, find a_1.

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2016 AIME I Problem 12: The first value with four prime factors

Find the least positive integer m for which $m^2 - m + 11$ has at least four prime factors, counted with multiplicity.

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2016 AIME II Problem 5: An infinite chain of right-triangle altitudes

Right triangle ABC_0 has pairwise coprime positive integer sides and perimeter p. Let C_1 be the foot from C_0 to AB; for $n \geq 2$, let C_n be the foot from C_{n-1} to $C_{n-2}B$. If $C_0C_1 + C_1C_2 + \cdots = 6p$, find p.

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2016 AIME II Problem 7: Three nested squares with two integer areas

Two concentric squares have parallel sides. The outer area is 2016 and the inner area is a smaller positive integer. A third square has one vertex on each outer side, while each inner vertex lies on one of its sides. Find the difference between the greatest and least possible positive integer areas of this third square.

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2016 AIME II Problem 8: Three distinct factors of a product

Count unordered triples of distinct positive integers whose product is $11 \cdot 21 \cdot 31 \cdot 41 \cdot 51 \cdot 61$.

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2016 AIME II Problem 9: An arithmetic sequence plus a geometric sequence

Increasing positive integer sequences satisfy $a_1 = b_1 = 1$, with a_n arithmetic and b_n geometric. Put $c_n = a_n + b_n$. If $c_{k-1} = 100$ and $c_{k+1} = 1000$, find c_k.

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2016 AIME II Problem 11: Divisor counts of perfect powers

Call N k-nice when some kth power has exactly N positive divisors. Count $1 \le N < 1000$ that are neither 7-nice nor 8-nice.

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2017 AIME I Problem 2: Equal remainders determine two divisors

Dividing 702,787,855 by a positive integer m gives the same positive remainder r. Dividing 412,722,815 by a positive integer n gives the same positive remainder ser. Find $m + n + r + s$.

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2017 AIME I Problem 3: A periodic sequence of triangular-number digits

Let d_n be the units digit of $1 + 2 + \cdots + n$. Find $\sum_{n=1}^{2017} d_n$ modulo 1000.

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2017 AIME I Problem 5: One rational number in two bases

A rational number is $(ab.cd)_8$ and also $(bb.ba)_{12}$, with all four digits nonzero. Find the decimal integer $100a + 10b + c$.

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2017 AIME I Problem 9: A concatenation-type recurrence modulo99

Let $a_{10} = 10$ and $a_n = 100a_{n-1} + n$ for integers $n > 10$. Find the least $n > 10$ with $99 \mid a_n$.

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2017 AIME I Problem 12: Product-free subsets

Count subsets S of $\{1, \ldots, 10\}$ containing no a, b, c (possibly equal) with $ab = c$.

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2017 AIME I Problem 13: The last gap between consecutive cubes

For integer $m \geq 2$, let $Q(m)$ be the least positive integer such that every integer $n \geq Q(m)$ admits a cube with $n < k^3 \leq mn$. Find $\sum_{m=2}^{2017} Q(m)$ modulo 1000.

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2017 AIME I Problem 14: Nested logarithms become a monotone power equation

For $a > 1, x > 1$, suppose $\log_a(\log_a(\log_a 2) + \log_a 24 - 128) = 128$ and $\log_a(\log_a x) = 256$. Find x modulo 1000.

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2017 AIME II Problem 4: Ternary digit avoidance

Count positive integers at most 2017 with no zero digit in base 3.

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2017 AIME II Problem 6: A square squeezed between nearby shifted squares

Find the sum of positive integers n for which $\sqrt{n^2 + 85n + 2017}$ is an integer.

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2017 AIME II Problem 8: Integrality of a truncated exponential series

Count positive integers $n < 2017$ for which $\sum_{j=0}^{6} n^j/j!$ is an integer.

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2018 AIME I Problem 2: A digit permutation in three bases

An integer is $(abc)_{14} = (acb)_{15} = (acac)_6$, with $a > 0$. Find its decimal value.

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2018 AIME I Problem 6: Two disjoint root-of-unity families

Count complex numbers z satisfying $|z| = 1$ and $z^{720} - z^{120} \in \mathbb{R}$. Find the count modulo 1000.

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2018 AIME I Problem 11: A two-digit ending requires prime-square orders

Find the least positive n such that the final two digits of 3^n in base 143 are 01.

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2018 AIME I Problem 12: Subset sums modulo three

Choose a subset of [18] uniformly. Find the numerator of its reduced probability of having sum divisible by 3.

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2018 AIME II Problem 2: A recurring triple proves periodicity

Let $a_0 = 2, a_1 = 5, a_2 = 8$, and for $n > 2$ let a_n be the least nonnegative residue of $4(a_{n-1} + a_{n-2} + a_{n-3})$ modulo 11. Find $a_{2018} a_{2020} a_{2022}$.

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2018 AIME II Problem 3: A square and a cube in the same base

Find the sum of integer bases $8 \leq b < 1000$ for which $(36)_b$ is a square and $(27)_b$ is a cube.

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2018 AIME II Problem 7: Rational lengths among equal-area parallel slices

A triangle has sides $AB = 9$, $AC = 12$, $BC = 5\sqrt{3}$. Lines parallel to BC divide it into 2450 equal-area regions. Let the kth parallel segment, including BC as the 2450 th, bound the first k regions nearest A. How many of these 2450 segments have rational length?

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2019 AIME I Problem 1: A sum of strings of nines

Let $N = 9 + 99 + 999 + \cdots + (10^{321} - 1)$. Find the sum of its decimal digits.

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2019 AIME I Problem 7: Prime exponents hidden in gcd and lcm logarithms

Positive integers x,y satisfy $\log_{10} x + 2 \log_{10} \gcd(x, y) = 60$ and $\log_{10} y + 2 \log_{10} \operatorname{lcm}(x, y) = 570$. If m,n count the prime factors of x,y with multiplicity, find $3m + 2n$.

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2019 AIME I Problem 9: Consecutive integers with seven divisors in total

Let $\tau(n)$ count positive divisors. Find the sum of the six least positive integers n such that $\tau(n) + \tau(n + 1) = 7$.

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2019 AIME I Problem 11: A circle tangent to all three excircles

An integer-sided triangle has $AB = AC$. A circle centered at its incenter is internally tangent to the A-excircle and externally tangent to the other two excircles. Find the smallest possible perimeter.

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2019 AIME I Problem 14: An eighth power forces order sixteen

Find the least odd prime divisor of $2019^8 + 1$.

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2019 AIME II Problem 4: A square product of four die rolls

Find numerator plus denominator of the probability that four fair die outcomes have square product.

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2019 AIME II Problem 9: Multiples of twenty having twenty divisors

Call a positive integer n k-pretty when it has exactly k positive divisors and is divisible by k. Let S be the sum of all 20-pretty integers below 2019. Find $S/20$.

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2019 AIME II Problem 10: A tangent sign itinerary fixes every binary digit

An angle $0 < \theta < 90^\circ$ has $\tan(2^n\theta) > 0$ precisely when $3 \mid n$, and negative otherwise, for every integer $n \geq 0$. Write $\theta = p/q$ degrees in lowest terms and find $p + q$.

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2019 AIME II Problem 12: Divisibility chains with prescribed sum

Count finite strictly increasing positive integer sequences, each term dividing the next, whose sum is 360.

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2019 AIME II Problem 14: A three-denomination Frobenius problem

With unlimited stamps of denominations $5, n, n+1$, the largest unattainable value is 91. Find the sum of possible n.

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2020 AIME I Problem 3: A shared set of digits in bases eleven and eight

A positive integer N has base-eleven digits abc and base-eight digits 1 bca. Find the least possible decimal N.

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2020 AIME I Problem 4: A truncated prefix must divide the suffix

Let S consist of positive integers ending in 2020 such that deleting those four digits leaves a positive divisor of the original integer. Find the sum of all decimal digits of all members of S.

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2020 AIME I Problem 9: A random chain of divisors

Independently sample three divisors a, b, c of 20^9 uniformly with replacement. Find the reduced numerator of $P(a \mid b \text{ and } b \mid c)$.

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2020 AIME I Problem 10: A missing prime power constrains exponential divisibility

Positive integers m,n satisfy $\gcd(m+n, 210) = 1$, $n^n \mid m^m$, and $n \nmid m$. Find the least possible $m+n$.

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2020 AIME I Problem 12: Prime-power divisibility after selecting an exponent class

Let n be the least positive integer such that $3^3 5^5 7^7 \mid 149^n - 2^n$. Find the number of positive divisors of n.

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2020 AIME II Problem 1: Square divisors of a prime-power product

Count ordered pairs of positive integers m,n with $m^2 n = 20^{20}$.

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2020 AIME II Problem 5: The smallest integer with nested digit-sum restrictions

Let f(n) be the base-four digit sum of n, and g(n) the base-eight digit sum of f(n). Let N be the least positive n for which the hexadecimal representation of g(n) uses a digit above 9. Find N modulo 1000.

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2020 AIME II Problem 10: Clear denominators before reducing a cube sum

Find the sum of all positive integers n for which $1^3 + \cdots + n^3$ leaves remainder 17 upon division by n+5.

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2021 AIME I Problem 3: Differences of powers of two

Count positive integers below 1000 expressible as a difference of integral powers of 2.

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2021 AIME I Problem 5: Integer arithmetic progressions satisfying a square identity

A strictly increasing three-term integer arithmetic progression is special when the sum of its squared terms equals its middle term times the square of its common difference. Find the sum of the final terms of all special progressions.

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2021 AIME I Problem 10: A rational recurrence reduces at its first shared prime

Start with $a_1 = 2020/2021$. When $a_k = m/n$ is reduced with positive integers m,n, define $a_{k+1} = (m+18)/(n+19)$. Find the sum of all j for which $a_j = t/(t+1)$ for a positive integer t.

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2021 AIME I Problem 14: A universal divisor-sum congruence without advanced prime distribution

Let sigma(a) be the sum of positive divisors of a. Let n be the least positive integer for which $2021 \mid \sigma(a^n) - 1$ for every positive integer a. Find the sum of the prime factors of n, counted with multiplicity.

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2021 AIME II Problem 3: Cyclic products modulo three

Permute $1, \ldots, 5$ around five labeled positions. Count permutations for which the sum of the five products of three cyclically consecutive entries is divisible by 3.

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2021 AIME II Problem 9: Counting exponent pairs by valuations

Count ordered $1 \le m, n \le 30$ with $\gcd(2^m + 1, 2^n - 1) > 1$.

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2021 AIME II Problem 13: Lift an exponential fixed point through powers of five

Find the least positive integer n with $1000 \mid 2^n + 5^n - n$.

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2021 AIME II Problem 15: Walking to the next square with two step sizes

For positive integers n, define f(n)=$\sqrt{n}$ at squares and f(n)=1+f(n+1) otherwise. Define g(n)=$\sqrt{n}$ at squares and g(n)=2+g(n+2) otherwise. Find the least n for which f(n)/g(n)=4/7.

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2022 AIME I Problem 2: A cyclic digit shift in two bases

The decimal integer with digits abc equals the base-nine integer with digits bca. Find the decimal integer.

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2022 AIME I Problem 7: A minimal positive ratio of digit products

Partition digits 1 through 9 into three ordered triples. Find numerator plus denominator of the least positive value $(abc - def)/(ghi)$.

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2022 AIME I Problem 13: Distinct reduced numerators

Consider all nonzero repeating four-digit decimals. Count their distinct reduced numerators modulo 1000.

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2022 AIME I Problem 14: Perimeter-bisecting cleavers determine a120-degree angle

A triangle has BC=219 and positive integer sides AB,AC. The lines through the midpoints of AB and AC that divide the triangle into two regions of equal perimeter meet at 30 degrees. Find its perimeter.

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2022 AIME II Problem 1: A population ratio constrains the total size

Adults were 5/12 of a crowd before a bus brought 50 people. They were 11/25 afterward. Find the least possible number of adults after the bus arrived.

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2022 AIME II Problem 5: Prime-difference triangles

Twenty cyclic vertices are labeled $1, \ldots, 20$; connect labels whose ordinary difference is prime. Count triangles with original vertices.

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2022 AIME II Problem 8: Identifiability from three floor values

Count positive integers $n \leq 600$ uniquely determined among positive integers by $\lfloor n/4 \rfloor, \lfloor n/5 \rfloor, \lfloor n/6 \rfloor$.

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2022 AIME II Problem 13: A coefficient reduced to stars and bars

Find coefficient x^{2022} in $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$.

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2022 AIME II Problem 14: Complete coverage by a finite stamp collection

For denominations $a < b < c$, let f be the minimum number of stamps, using every denomination, whose subcollections represent every value from 1 through 1000. Find and sum the three least values of c permitting $f = 97$.

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2023 AIME I Problem 4: Sum all divisors with prescribed exponent parity

Sum all positive integers m for which 13!/m is a perfect square. If the sum is $2^a3^b5^c7^d11^e13^f$, find a+b+c+d+e+f.

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2023 AIME I Problem 7: Pairwise-distinct remainders

Count positive $n < 1000$ whose remainders modulo $2, 3, 4, 5, 6$ are all different.

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2023 AIME I Problem 10: Bound the floor error before summing residues

For a positive integer a, put $U = \sum_{n=1}^{2023} \lfloor (n^2 - na)/5 \rfloor$. Exactly one a makes −1000<U<1000. Find a+U.

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2023 AIME I Problem 15: A bounded Gaussian-integer search with all sign cases

Find the largest prime p<1000 for which a Gaussian integer z has absolute value $\sqrt{p}$ and p,Re(z cubed),Im(z cubed) are positive side lengths of a triangle.

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2023 AIME II Problem 2: Enumerate the four-digit octal palindromes

Find the greatest positive integer below 1000 that is palindromic in both decimal and base 8.

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2023 AIME II Problem 5: The cancellation divisor determines a rational number

Let S consist of positive rationals r for which r and 55 r, each reduced, have the same numerator-plus-denominator sum. Write the sum of S as p/q reduced; find p+q.

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2023 AIME II Problem 10: Balanced proper colorings of a ladder

Place $1, \ldots, 12$ in a 2 by 6 grid so adjacent cells never have equal residues modulo 3. Let N count placements. Find the number of positive divisors of N.

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2023 AIME II Problem 13: Prove the integrality class before counting last digits

For acute A, tan A=2 cos A. Count positive n≤1000 for which sec to the nth power A plus tan to the nth power A is an integer ending in 9.

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2023 AIME II Problem 15: A finite carry state detects repeated modular inverses

For each n≥1, let a_n be the smallest positive multiple of 23 congruent to 1 modulo 2 to the nth power. Count n≤1000 for which a_n=a_(n+1).

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2024 AIME I Problem 13: Lift the four roots of minus one

Let p be the least prime for which some positive n satisfies p squared divides n to the fourth power plus 1. Find the least positive m satisfying the same divisibility for that p.

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2024 AIME II Problem 2: A mode and a missing median constrain an integer partition

A finite list of positive integers sums to 30, has unique mode 9, and has an integral median that is absent from the list. Find the sum of squared entries.

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2024 AIME II Problem 6: Encoding a set by maxima of subsets

A finite set A of positive integers has the property that exactly 2024 finite nonempty sets of positive integers have maximum in A. Find the sum of the elements of A.

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2024 AIME II Problem 7: Sum digit replacements before solving the digits

Let N be the greatest four-digit integer such that replacing any one digit by 1 produces a multiple of 7. If N=1000 Q+R with 0≤R<1000, find Q+R.

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2024 AIME II Problem 13: Evaluate a root product at two conjugate constants

Let omega≠1 satisfy omega to the 13 th power=1. Find the product of $2-2\omega^k+\omega^{2k}$ for k=0 through 12, modulo 1000.

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2024 AIME II Problem 14: Digit-square conditions count CRT idempotents

For integer b≥2, call a positive integer special if its base-b representation has exactly two digits whose sum equals its square root. Find the least b with more than ten special integers.

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2025 AIME I Problem 1: Reduce a base-divisibility condition to one fixed integer

Find the sum of integer bases b>9 for which the base-b number 17 divides the base-b number 97.

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2025 AIME I Problem 4: Count a union of two primitive lattice lines

Count ordered integer pairs (x,y) with $-100\le x,y\le 100$ satisfying $12x^2-xy-6y^2=0$.

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2025 AIME I Problem 5: Digit permutations divisible by twenty-two

Let N count permutations of digits 1 through 8 that form an eight-digit integer divisible by 22. Find $N-2025$.

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2025 AIME I Problem 15: Cubic residues and three-variable counting

Count ordered triples $1 \le a, b, c \le 729$ with $a^3 + b^3 + c^3$ divisible by 2187, and give the remainder modulo 1000.

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2025 AIME II Problem 2: Evaluate a polynomial at the modulus root

Find the sum of positive integers n for which n+2 divides $3(n+3)(n^2+9)$.

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2025 AIME II Problem 7: The least common multiple of a random divisor subset

Choose uniformly a subset of the positive divisors of 2025. Find numerator plus denominator of the probability that its least common multiple is 2025.

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2025 AIME II Problem 8: When greedy coin change succeeds

With coin values $1, 10$, and 25, count amounts from 1 through 1000 for which repeatedly taking the largest affordable coin uses the fewest coins.

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2025 AIME II Problem 13: Track reduction before simplifying a rational recurrence

Define $x_1 = 25/11$ and $x_{k+1} = (x_k + 1/x_k - 1)/3$. If x 2025=m/n reduced with positive integers m,n, find (m+n) modulo 1000.

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2026 AIME I Problem 4: Products of two unequal factors

How many integers at most 100 have the form $a + b + ab$ for distinct positive integers a, b?

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2026 AIME I Problem 6: A logarithmic quadratic determines a product

Let P be the product of all positive real solutions of $\sqrt[20]{x^{\log_{2026} x}} = 26x$. Find the number of positive divisors of P.

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2026 AIME I Problem 8: Divisor residues determined by exponent parity

Count divisors of 17017^{17} congruent to 5 modulo 12, and give the count modulo 1000.

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2026 AIME I Problem 13: Binomial residue sums over a finite field

For $0 \le r < 502$, let S_r sum $\binom{10000}{k}$ over $0 \le k \le 10000$ congruent to r modulo 502. How many S_r are divisible by 503?

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2026 AIME II Problem 1: Arithmetic progressions indexed by common divisors

Find the sum of the eleventh terms of all integer arithmetic sequences beginning with 4 and containing both 24 and 34. An alternate archived version asks for the tenth terms instead.

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2026 AIME II Problem 3: An indented pentagon gives one residue class

A simple nonconvex pentagon $ABCDE$ has interior angles $A = E = 90°$, $B = D = 45°$, with $DE < AB$, $AE = 20$, $BC = 14\sqrt{2}$. Vertices B, C, D lie on one side of AE. How many integer lengths $AB < 2026$ make its area a multiple of 16?

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2026 AIME II Problem 4: A numeral equal to its value in a smaller base

For positive decimal n, reinterpret its digit string in the base one larger than its greatest digit. Count $n < 1000$ unchanged by this operation.

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2026 AIME II Problem 5: Equal hypergeometric probabilities

An urn has at least seven marbles of each of two colors. Drawing seven without replacement gives equal probabilities of exactly four red marbles and exactly five red marbles. Sum the five smallest possible total populations.

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2026 AIME II Problem 8: An integer isosceles triangle with a tiny incenter triangle

In an isosceles triangle ABC, $AB = BC$ and I is the incenter. The perimeters of ABC and AIC have ratio $125 : 6$. All side lengths of both triangles are integers. Find the least possible AB.

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2026 AIME II Problem 9: A divisor-generating decimal with controlled carries

Let $S = \sum_{k \ge 1} 1/(10^k - 1)$. Find $\lfloor 10^{100} S \rfloor$ modulo 1000.

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2026 AIME II Problem 10: An angle bisector and tangent secants

In triangle ABC, $AB = 200$, $AC = 225$, and AD bisects angle A, with D on BC. A circle through A is tangent to BC at D and meets AB, AC again at E, F on those segments. If AE, AF, BD, CD are positive integers, find the greatest possible BC. An alternate version asks for the sum of all possible values.

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2026 AIME II Problem 15: A Fano-plane residue count

Count ordered seven-tuples with entries in $\{1, 2, 3\}$ for which both $\sum_i a_i$ and $\sum_i a_i a_{i+1} a_{i+3}$ are divisible by three; indices are cyclic modulo seven.

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